(Please write your Exam Roll No.)

Roll No........................

First Term Examination

Sixth Semester [B.TECH]

Time: 1: 30 hrs **February -2017**

Paper Code: ETCS-310 **Max Marks: 30**

Subject: Artificial Intelligence

Note: Attempt any three questions including Q.No. 1 which is compulsory.

Q1.

(2*5)

a) How many categories do intelligent System fall into? Briefly explain their horizon too?

b) Explain knowledge based system in context of Artificial Intelligence.

c) Write the pseudocode program for Model Based Reflex Agent.

d) Construct the truth table for :

$A = (P \vee Q) \Rightarrow ((P \vee R) \Rightarrow (R \vee Q))$

e) What do you mean by free and bound variable? Identify free and bound variable in the following wff :

$\forall x\, (\exists y\, P(x, y) \wedge \forall x\, (Q(x) \vee R(x) \Rightarrow T(x, y)$

Q2. a) Solve the problem of a simple household robot of moving a desk from one room to other with two things on it using Means End analysis. The operators available are: PUSH, CARRY, WALK, PICKUP, PUTDOWN, PLACE. (5)

b) Differentiate between Uninformed and Informed Search technique with the help of example. (5)

Q3.

a) Consider the following axiom:

i) P ii) $(P \wedge Q) \Rightarrow R$ iii) $(S \vee T) \Rightarrow Q$ iv) T

Prove that R is true by Resolution. (5)

b) Differentiate between propositional logic and First order logic with the help of example. (5)

Q4.

a) Trace the constraint satisfaction procedure for solving the following cryptarithmetic

Problem:

```
    FOUR
 +  MICE
 -------
   FOUND
```

(6)

b) Check the validity of the following argument by applying rule of inference:" If the races are fixed then the casinos are crooked, then the tourist trade will decline. If the tourist trade declined, then the police will be happy. The police force never be happy. Therefore the races are not fixed". (4)

CS/19

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Exam Roll No.:..................

MID TERM EXAMINATION

SIXTH SEMESTER [B.TECH.] FEB-2019

Paper Code: ETCS-310

Subject: Artificial Intelligence

Time: 1:30 Hours

Maximum Marks: 30

Note: Attempt any Three questions including Q. No. 1, which is compulsory

Q No. 1

a) How do you define Artificial Intelligence in terms of its task domain?

b) How production system useful for solving AI problems?

c) What approaches required for the representation of knowledge in a particular domain?

d) A problem-solving search can proceed either forward or backward. What factors determine the choice of direction for a particular problem?

[2.5X4]

Q No. 2

a) What is AI? How do you compare human intelligence with artificial intelligence? [5]

b) How do you define the characteristics of a problem, for the solution that is desired and the circumstances under which the solution must take place? [5]

Q No. 3

a) Consider trying to solve the 8-puzzla using hill climbing. Can you find a heuristic function that makes this work? Make sure it works on the following example:

1	2	3
8	5	6
4	7	

1	2	3
4	5	6
7	8	

[5]

Q No. 4

a) Show that how means-ends analysis could be used to solve the problem of getting from one place to another. Assume that the available operators are walk, drive, take the bus, take a cup, and fly. [5]

b) Assume the following facts:

- Mohen only likes easy courses.
- Science courses are hard.
- All courses in Humanities are easy.
- HS-301 is a Humanities course.

Use resolution to answer the question, "What course would Mohen Like?"

[5]

END

CSE/IT

(Please Write your Roll No. immediately)

Roll No ....

Mid-Term Examination

B.Tech – VI Semester
Paper Code: ETCS-310
Time: 1.5 hour

April, 2022
Subject – Artificial Intelligence
Max Marks: 30

Note: Q.No. 1 is Compulsory. Attempt any two more Question from the rest.

Q1. **(2*5=10 marks)**

a.) Differentiate Propositional logic and FOPL with example.

b.) What is PEAS? Specify PEAS for an Aerospace System.

c.) Obtain disjunctive normal form of:

$p \vee (\sim p \Rightarrow (q \vee (q \Rightarrow \sim r)))$

d.) Explain Turing Test. Why Turing Test was criticised and also explain Chinese room Argument Test.

e.) Differentiate OR graph and AND-OR graph.

Q2. a.) What is heuristic search? Explain Hill Climbing and its limitations. **(5 marks)**

b.) Explain Iterative Deepening Depth First Search with example. **(5 marks)**

Q3. a.) Represent the following facts in FOPL: **(5 marks)**

i.) All men are mortal

ii.) Some pet dogs are dangerous

iii.) All basketball players are tall

iv.) Lipton is a tea

v.) Some employees are sick today.

b.) Check the validity of following argument: **(5 marks)**

"If I get the job and work hard, then I will get promoted. If I get promoted, then I will be happy. I will not be happy. Therefore, either I will not get the job or I will not work hard".

Q4. a.) Solve following cryptographic puzzle using Constraint Satisfaction procedure: **(5 marks)**

```
   FOUR
 + MICE
 ------
  FOUND
```

a.) Discuss forward and backward chaining with suitable example. **(5 marks)**

CS/11
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(Please write your Exam Roll No.)

Roll No.........................

First Term Examination

Sixth Semester [B.TECH]

Time: 1: 30 hrs

February -2016

Paper Code: ETCS-310

Max Marks: 30

Subject: Artificial Intelligence

Note: Attempt any three questions including Q.No. 1 which is compulsory.

Q1.

(2X5)

a) Define a) Intelligence b) Artificial Intelligence c) Agent d) Logical Reasoning.
b) What do you mean by PEAS? Specify PEAS for Medical diagnosis system.
c) Differentiate between Declarative and procedural Knowledge with example.
d) Formulate Predicate logic statement for "All cats like fish, cats eat everything they like, and Tom is a cat" .Does Tom eat fish?
e) What is a Skolem function? Explain with example.

Q2.

a) What do you mean by Hill Climbing search technique? What are various problem encountered in it? (5)
b) Write an algorithm for Best First Search algorithm. (5)

Q3.

a) Consider the following axiom: (5)
   i. Anyone whom Mary likes is a football star.
   ii. Any student who doesn't pass doesn't play.
   iii. John is a student who does not study doesn't pass.
   iv. Anyone who doesn't play is not a football star.
   Prove that if John doesn't study, then Mary doesn't likes John.
b) Differentiate between forward chaining and backward chaining with the help of example. (5)

Q4. a) Trace the constraint satisfaction procedure for solving the following cryptarithmetic

Problem:

```
   B R O W N
 + Y E L L O W
 -------------
   P U R P L E
```

(6)

b) What are the properties of propositional calculus, explain with example. (4)

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(Please write your Exam Roll No.)

Exam Roll No.:....................

MID TERM EXAMINATION

SIXTH SEMESTER [B.TECH.] FEB-2020

Paper Code: ETCS-310

Subject: Artificial Intelligence

Time: 1:30 Hours **Maximum Marks: 30**

Note: Attempt any Three questions including Q. No. 1, which is compulsory

Q No. 1

a) What is a heuristic function? Give an example.

b) What are common AI techniques used in all application domains?

c) What approaches required for the representation of knowledge in a particular domain?

d) Suggest an application domain where backward chaining is suitable?

[2.5X4]

Q No. 2

a) What analysis is required for choosing a good search strategy?

[5]

b) You are given two jugs with measuring marks, a 4-gallon one and a 3-gallon one. There is a pump to fill the jugs with jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? For the agent of water jug, develop a state space description.

[5]

Q No. 3

a) Consider trying to solve the 8-puzzla using hill climbing. Can you find a heuristic function that makes this work? Make sure it works on the following example:

1	2	3
8	5	6
4	7	

1	2	3
4	5	6
7	8	

[5]

Q No. 4

a) Show that how means-ends analysis could be used to solve the problem of getting from one place to another. Explain with the help of an example.

[5]

b) Assume the following facts:

- Mohan only likes easy courses.
- Science courses are hard.
- All courses in Humanities are easy.
- HS-301 is a Humanities course.

Use resolution to answer the question, "What course would Mohan Like?"

[5]

END

(Please write your Roll No. Immediately)

Roll No.................................

FIRST TERM EXAMINATION
SIXTH SEMESTER B.TECH. (February 2018)

Paper Code: **ETCS-310(CSE/IT)**
Time: **1 and 1/2 Hour**

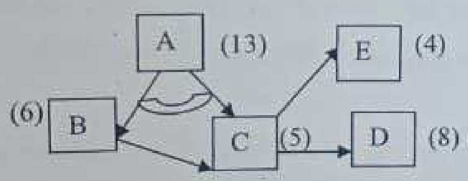
Sub: **Artificial Intelligence**
Max. Marks : **30**

Note: **Question No. 1** is compulsory, Attempt any 2 questions from the rest.

Q1. a) Write PEAS descriptions for Medical Diagnosis System.
b) What is a heuristic function? Explain its components.
c) Compare the time complexity of Breadth first search and Iterative deepening depth first search.
d) Give examples of Decomposable and Predictable problems.
e) Differentiate between OR graph and AND-OR graph. [2X5=10]

Q2. a) What are the problems of Hill Climbing algorithm? In what ways they can be dealt with. [3]
b) Write the algorithm of Means Ends Analysis. [4]
c) Describe Learning Agent. [3]

Q3. a) Explain differences between Propositional logic and First order predicate logic. [2]
b) Show the backward cost propagation of AO* algorithm of the following graph if the cost of node D changed from 8 to 3 now. [4]



c) Represent the following facts in Predicate Logic:
(i) Bill dislikes children who eat chocolate.
(ii) If it is raining you should take an umbrella.
(iii) All people who drinks milk either thinks milk is healthy or hates people who drinks tea. [4]

Q4. a) Trace the Constraint Satisfaction procedure solving the following crypto arithmetic problem.

$$\begin{array}{r} S\,A\,V\,E \\ T\,R\,E\,E \\ \hline S\,A\,F\,A\,R \end{array}$$ [6]

b) Consider the following statement
i) XYZ is a serial in Sony
ii) Ruma likes all serials in ZEE
iii) Anyone who likes serials in ZEE does not like serials in Sony.
Prove by Backward Reasoning that Ruma does not like serial XYZ. [4]

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